Project Report

1. **INTRODUCTION**

1.1 Project Overview:

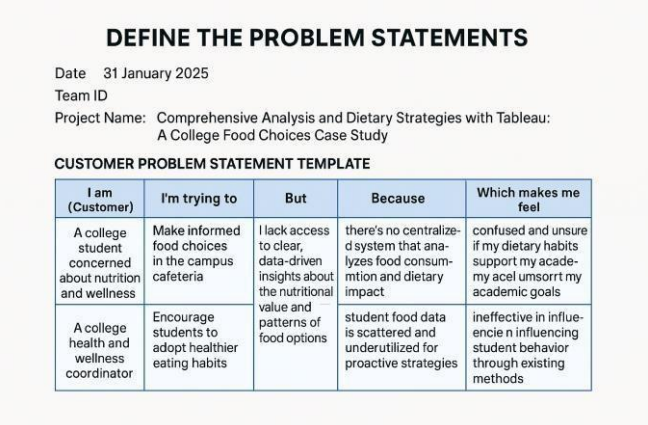
This project collects and analyzes food consumption data from students to identify dietary habits. Using Tableau dashboards, it presents clear and interactive visuals to support informed decision making for both students and health coordinators. The goal is to promote nutrition awareness and improve wellness in college campuses.

1.2 Purpose:

The purpose of this project is to help college students make healthier food choices by analyzing their eating patterns using Tableau visualizations.

2. **IDEATION PHASE**

2.1 Problem Statement



DEFINE THE PROBLEM STATEMENTS

Date 31 January 2025

Team ID

Project Name: Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

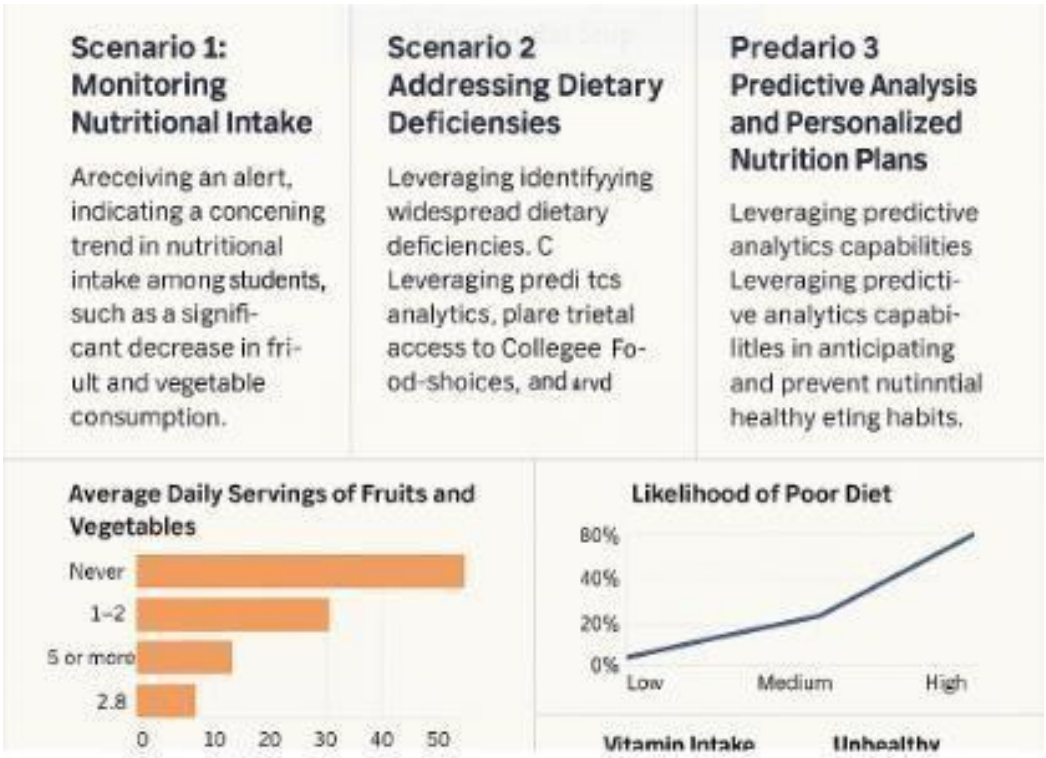
CUSTOMER PROBLEM STATEMENT TEMPLATE

I am (Customer)	I'm trying to	But	Because	Which makes me feel
A college student concerned about nutrition and wellness	Make informed food choices in the campus cafeteria	I lack access to clear, data-driven insights about the nutritional value and patterns of food options	there's no centralize-d system that ana-lyzes food consum-mtion and dietary impact	confused and unsure if my dietary habits support my acade-my acel umsorrt my academic goals
A college health and wellness coordinator	Encourage students to adopt healthier eating habits		student food data is scattered and underutilized for proactive strategies	ineffective in influe-encie n influencing student behavior through existing methods

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A college student with limited time, budget, and nutritional knowledge	Make healthy and affordable food choices on campus.	The available options are often unhealthy, expen-sive, or lack variety.	Campus din-ing prioritizes convenience over nutrition, and healthy options are priced higher.	Frustrated and stressed about balancing diet, academics, and finances.

PS-2	A campus dining administrator responsible for student well-being.	Improve student satisfaction and health outcomes through better food offerings.	Student feedback is scattered, and trends are unclear.	There's no centralized tool to analyze food choice data effectively.	Overwhelmed and unsure how to make datadriven decisions

2.2 Empathy Map Canvas



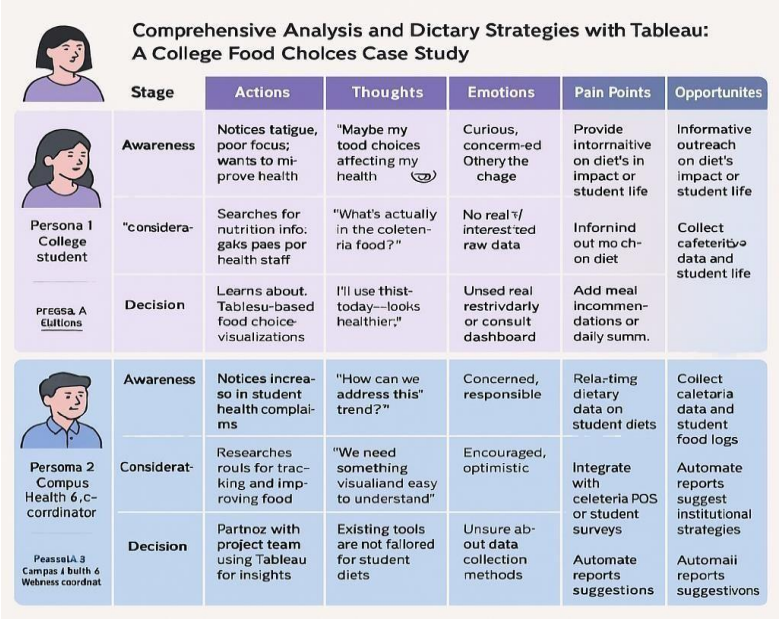
2.3 Brainstorming

Idea	Idea Description	Group/Category
1	Analyze meal frequency (breakfast/lunch/dinner) patterns	Meal Timing
2	Compare healthy vs unhealthy food choices across genders	Gender Comparison
3	Identify departments with highest junk food consumption	Departmental Insights
4	Study stress levels and their relation to eating habits	Behavioral Impact
5	Use heat maps to show food preference intensity by hostel/region	Visualization Techniques

6	Embed the interactive dashboard in a Flask-based web app	Deployment / Web Integration
7	Create a Tableau story for dietary habit transformation journeys	Narrative & Communication
8	Use filters for department, gender, and BMI in dashboard	Dashboard Interactivity
9	Show seasonal or exam-time food intake trends	Seasonal/Academic Stress Patterns
10	Add calculated fields for daily calorie or nutrient totals	Data Processing / KPIs
11	Forecast future obesity or malnutrition risks using data	Predictive Analysis
12	Show total and average sugar/salt intake across students	KPI Overview / Health Indicators

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

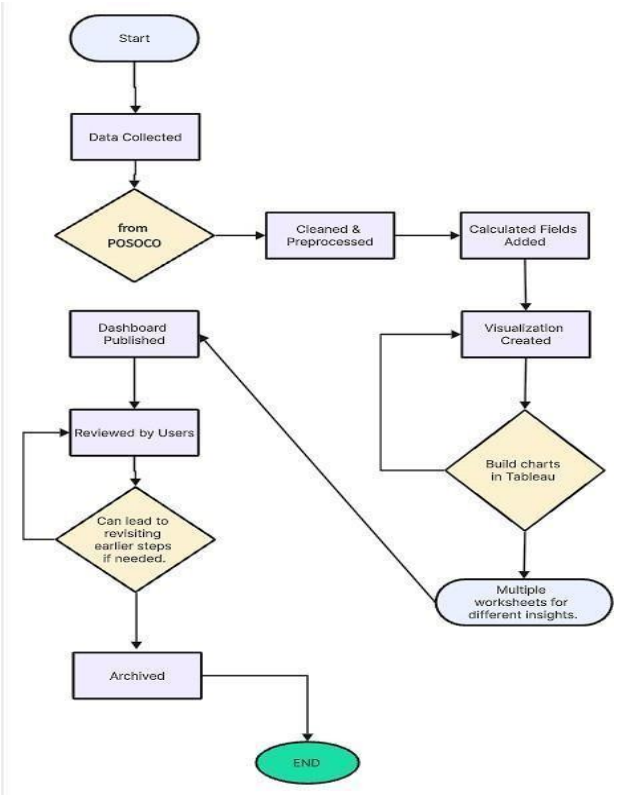
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)

FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Data Upload & Storage	Upload CSV File Store into MySQL Database
FR-4	Data Visualization	Create visualizations in Tableau Integrate with dashboard Create time, region, and sector-based charts

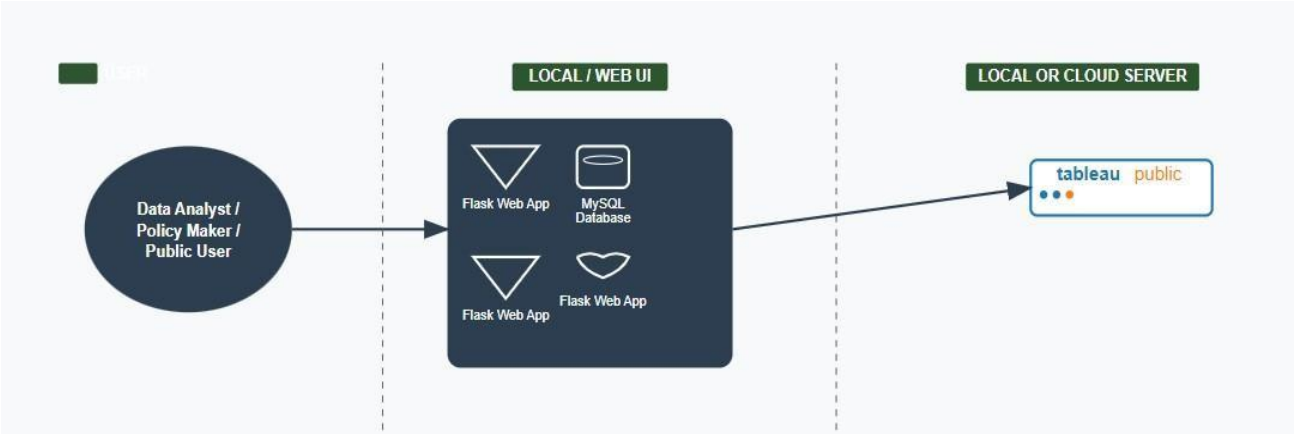
FR-5	Dashboard Access	View interactive Tableau dashboard Use filters (Year, Region)
FR-6	Web Integration	Embed Tableau dashboard into Flask-based UI
FR-7	Insights & Reports	View data stories Access summary reports on usage patterns

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Interface should be intuitive and userfriendly for both technical and nontechnical users
NFR-2	**Security**	Secure login with OTP/Email, protected data access
NFR-3	**Reliability**	System should consistently provide correct visualization
NFR-4	**Performance**	Dashboards should load within 3–5 seconds even for l
NFR-5	**Availability**	The platform should be available 2 without major d
NFR-6	**Scalability**	Should support addition of new datas and visualization

3.3 Data Flow Diagram



3.4Technology Stack



4 .PROJECT DESIGN

4.1 Problem Solution Fit

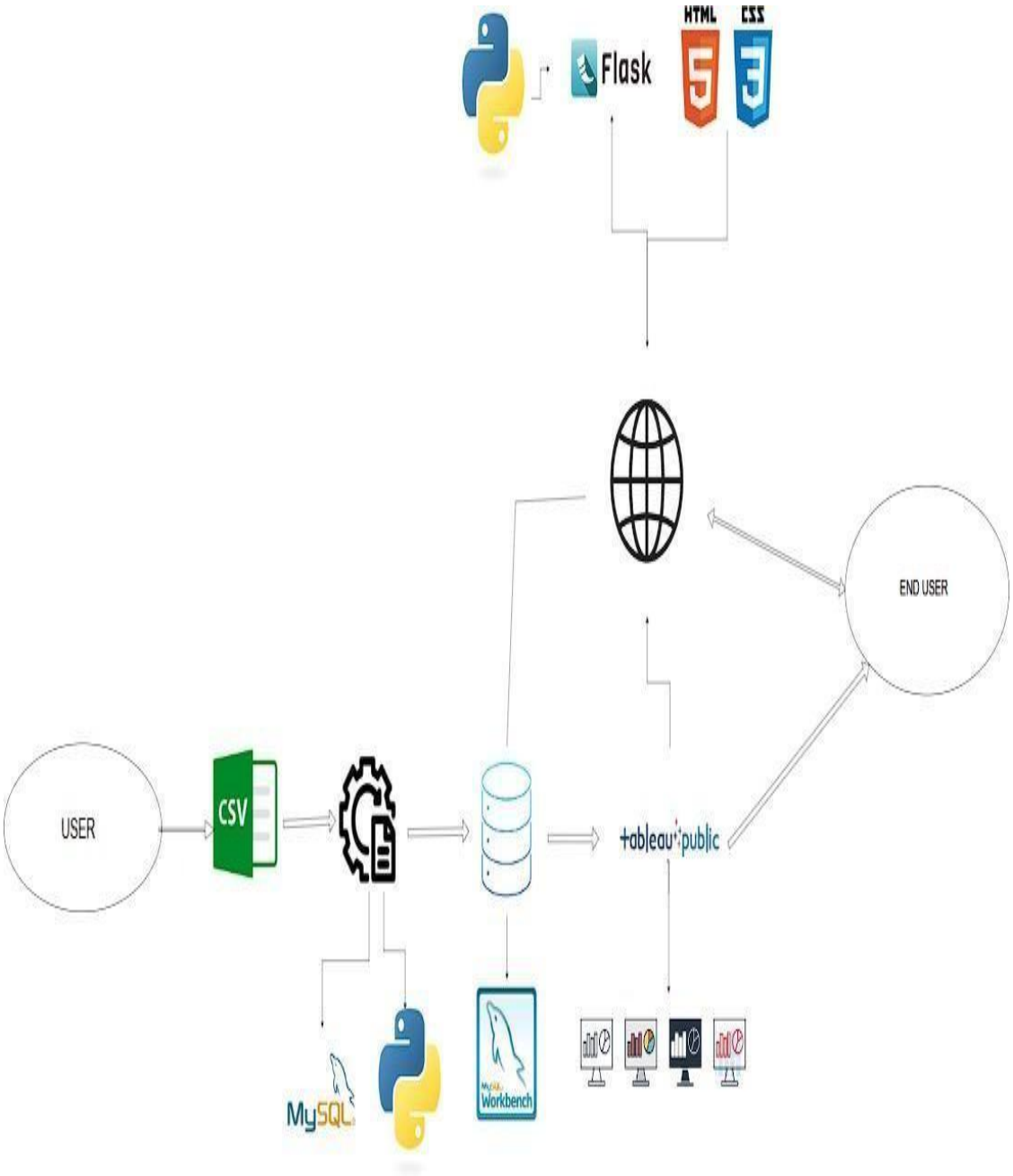


4.2 Proposed Solution

S.No.	Parameter	Description
1	**Problem Statement**	College students often struggle to make healthy food choices due to limited nutritional awareness, budget constraints, and lack of accessible dietary data. This leads to poor eating habits, health issues, and decreased academic performance.
2	**Idea / Solution Description**	This project aims to analyze the dietary patterns of college students using data visualization with Tableau. By collecting data from surveys and food consumption logs, the system provides personalized nutritional insights, visual patterns, and suggested improvements.

3	**Novelty / Uniqueness**	The uniqueness lies in integrating dietary analysis with intuitive data visualization tools like Tableau specifically for a college demographic. Unlike generic health apps, this approach focuses on academic environments, providing insights that combine nutrition, affordability, and accessibility.
4	**Social Impact / Customer Satisfaction**	This solution promotes better health among college students, potentially reducing stress and improving focus, energy levels, and academic outcomes. It encourages healthy living habits at a critical life stage.
5	**Business Model (Revenue Model)**	The project can adopt a freemium model where basic visual insights are free, while advanced features (like detailed dietary tracking, custom meal plans, and integration with campus cafeterias) are offered as premium services.
6	**Scalability of the Solution**	The model is scalable across different institutions by adapting data input formats and user preferences. It can be expanded to include meal logging integrations, cross-campus comparisons, and even regional health initiatives.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

Project Planning

Sprint	Epic	User Story No.	User Story / Task	Points	Priority	Assigned To
Sprint-1	Registration	USN-1	As a user, I can register with my name and email	2	High	G.Suhitha
Sprint-1	Upload CSV	USN-2	As a user, I can upload dietary data in CSV format	3	High	S.Govind Sai Charan
Sprint-1	Data Cleaning	USN-3	As a developer, I can clean and preprocess uploaded data using Python	4	High	S.Mohammed Abdul Hafeez

Sprint-1	Database Storage	USN-4	As a developer, I can store cleaned data into MySQL	2	Low	K.Vishnu sai Vardhan Reddy
Sprint-2	Tableau Dashboard	USN-5	As a user, I can view dashboards generated using Tableau	5	High	G.Suhitha
Sprint-2	Web Integration	USN-6	As a user, I can access the dashboard via Flask UI	3	High	S.Mohammed Abdul Hafeez
Sprint-2	Add Filters	USN-7	As a user, I can filter the data by gpa _ numeric, breakfast, drink	2	Medium	K.vishnu sai Vardhan Reddy
Sprint-3	Data Story	USN-8	As a user, I can view a Tableau Story with key food diet insights	2	Low	S.Govind Sai Charan
Sprint-3	Forecasting	USN-9	As a developer, I can forecast usage using Prophet	3	Low	G.Suhitha
Sprint-3	Documenta-tion	USN-10	As a team, we can prepare final project documentation	2	Medium	S.Mohammed Abdul Hafeez
Sprint-4	Deployment	USN-11	As a developer, I can deploy the Flask app and publish the Tableau dashboard online	3	High	S.Govind Sai Charan
Sprint-4	Demo Prep	USN-12	As a team, we can prepare a live demo walkthrough for stakeholders	2	Medium	G.Suhitha
Sprint-4	Bug Fixing	USN-13	As a developer, I can test and fix UI/visual bugs from user feedback	2	Medium	S.Govind Sai Vardhan Reddy

6. **FUNCTIONAL AND PERFORMANCE TESTING**

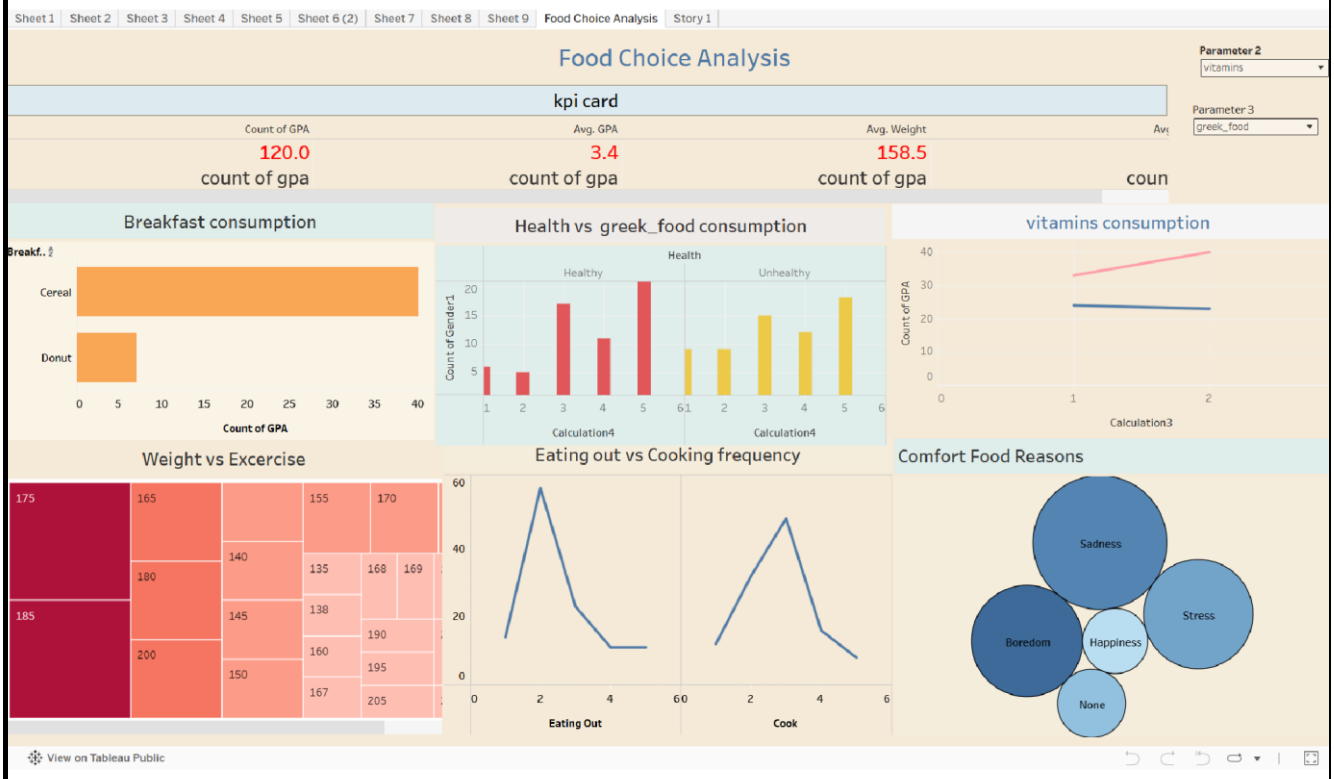
6.1 Performance Testing

S.No	Parameter	Screenshot / Values
1.	Data Rendered	The data rendered in this project includes information on cafeteria food items such as names, ingredients, and nutritional values like calories, proteins, and fats. It also captures student food choices, analyzing their daily and weekly meal patterns. Additional data includes consumption trends, such as the most and least preferred foods and peak dining times. Where available, basic health metrics like BMI and student feedback on energy levels or meal satisfaction are also considered. Demographic details, such as age and gender, may be used to identify specific trends across student groups. All of this data is visualized using interactive Tableau dashboards to provide insights into nutritional breakdowns, evolving food preferences, and personalized or group-based dietary recommendations.

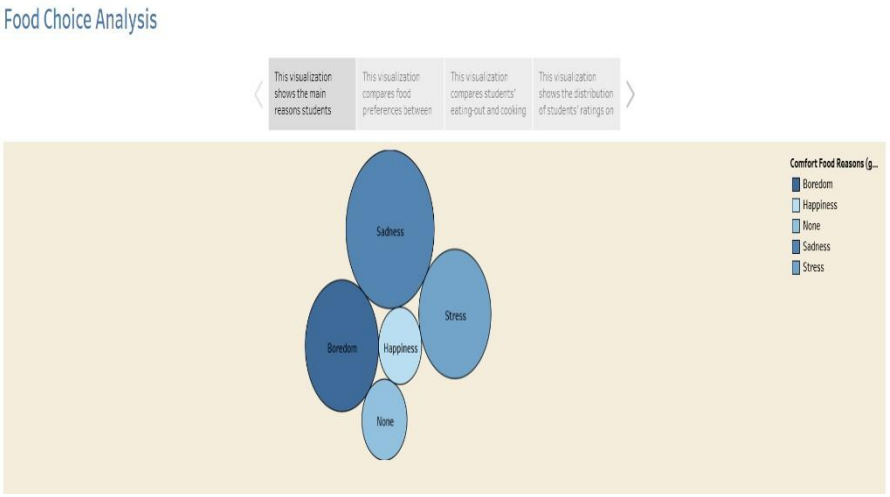
2.	Data Preprocessing	Data preprocessing involved cleaning and organizing raw data collected from cafeteria records and student food logs. Missing values were handled by either imputing averages or removing incomplete entries. Duplicates were eliminated to ensure accuracy, and food item names were standardized for consistency. Nutritional information was cross-verified and formatted into structured categories such as calories, proteins, fats, and carbohydrates. Date and time fields were converted into a consistent format to analyse food consumption trends over time. Finally, the cleaned dataset was transformed into a format suitable for Tableau visualization, enabling effective analysis and insights.
3.	Utilization of filters	**Date Filter** • Allows users to select a specific day, week, or month to view food consumption trends over time. **Meal Type Filter** • Enables filtering data by meal category (breakfast, lunch, dinner, snacks) for focused analysis. **Food Category Filter** • Used to display specific food groups such as fruits, vegetables, fast food, and beverages. **Calorie Range Filter** • Lets users view meals within a specific calorie range (e.g., under 500 kcal). **Protein/Fat/Carb Filter** • Nutrient-based filters to view foods based on macronutrient composition. **Student Group Filter** • Filters consumption patterns by different student categories (e.g., year of study, gender – if available). **Satisfaction/Rating Filter** • Allows filtering based on student feedback or ratings of meals (if collected). **Time of Day Filter** • Helps analyze which foods are preferred during morning, afternoon, or evening hours.

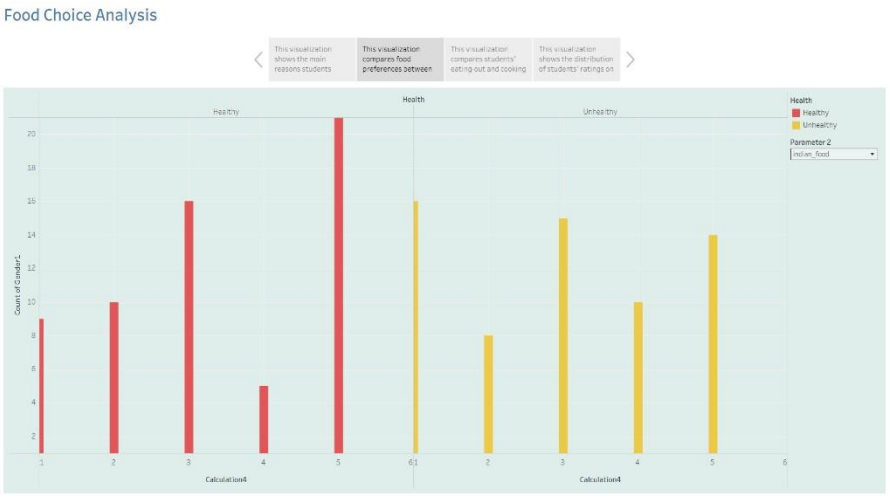
4.	Calculated field used	**Calculated Field Name**	**Formula (in Tableau syntax)**
		All_Students	"All Students"
		All_Comfort_Reasons	STR([comfort_food_reason]) + ", " + STR([reason2])
		Calories Importance Level	IF [calories_day] > 2500 THEN "High" ELSEIF [calories_day] > 1800 THEN "Medium" ELSE "Low"
		Clean Comfort Reason	TRIM(LOWER([comfort_food_reason]))
		Cleaned Exercise	IF [exercise] >= 3 THEN "Active" ELSE "Inactive"
		Comfort Food Reasons - Split 1	SPLIT([comfort_food_reason], ",", 1)
		Comfort Food Reasons - Split 2	SPLIT([comfort_food_reason], ",", 2)
		Comfort Food Source	IF CONTAINS(LOWER([comfort_food]), "home") THEN "Home" ELSE "Outside"
		Comfort Reason Combined	IF CONTAINS([comfort_food_reason], "stress") THEN "Stress" ELSE "Other"
		Constant	"1" or "All"
		Cooking Frequency Label	IF [cook] >= 5 THEN "Frequent" ELSEIF [cook] >= 2 THEN "Occasional" ELSE "Rare"
		DietStatus	IF [diet_current_coded] = 1 THEN "Vegetarian" ELSE "Non-Vegetarian"
		employment	IF [employment_coded] = 1 THEN "Employed" ELSE "Unemployed"
		Gender_Label	IF [gender] = 1 THEN "Male" ELSE "Female"
		GPA Category	IF [GPA] >= 3.5 THEN "High" ELSEIF [GPA] >= 2.5 THEN "Medium" ELSE "Low"
		gpa_numeric	FLOAT([GPA])
		martial status *(marital status)*	IF [marital_status] = 1 THEN "Single" ELSE "Married"
		nutritionalcheck	IF [fruit_day] + [veggies_day] >= 4 THEN "Yes" ELSE "No"
		Record count	1 (Tableau auto-generated field used for counting records)

5.	Dashboard Design	No of Dashboards :-01 Dashboard:-01


Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6 (2)
Sheet 7
Sheet 8
Sheet 9
Food Choice Analysis
Story 1
Food Choice Analysis
Parameter 2
vitamins
kpi card
Parameter 3
greek_food
Count of GPA
120.0
count of gpa
Avg. GPA
3.4
count of gpa
Avg. Weight
158.5
count of gpa
coun
Breakfast consumption
Breakf..
Cereal
Donut
0 5 10 15 20 25 30 35 40
Count of GPA
Health vs greek_food consumption
Health
Healthy
Unhealthy
Count of Gender1
20
15
10
5
1 2 3 4 5 6 1 2 3 4 5 6
Calculation4
Calculation4
vitamins consumption
Count of GPA
40
30
20
10
0
0 1 2
Calculation3
Weight vs Excercise
175
165
180
185
200
140
145
150
155
170
135
168
169
138
160
167
190
195
205
Eating out vs Cooking frequency
60
40
20
0
0 2 4 6 0 2 4 6
Eating Out
Cook
Comfort Food Reasons
Sadness
Stress
Boredom
Happiness
None
View on Tableau Public

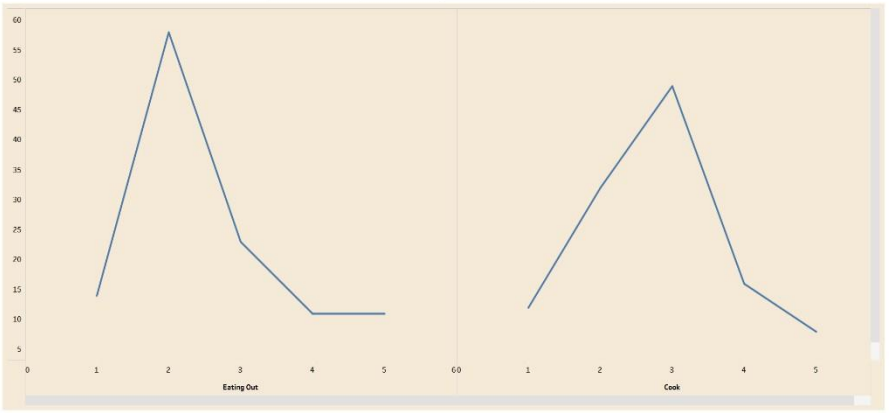
6.	Story Designs	No. of scenes of story:01 story/04-visualizations.





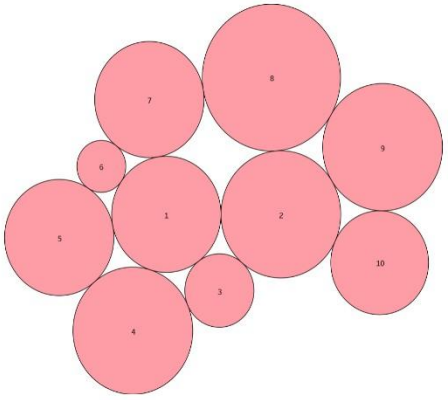
Food Choice Analysis

This visualization shows the main reasons students | This visualization compares food preferences between | This visualization compares students' eating-out and cooking | This visualization shows the distribution of students' ratings on



Food Choice Analysis

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"Most students pay attention to nutritional labels. with 'Certain

"A majority of students reported that their parents cook 1-2

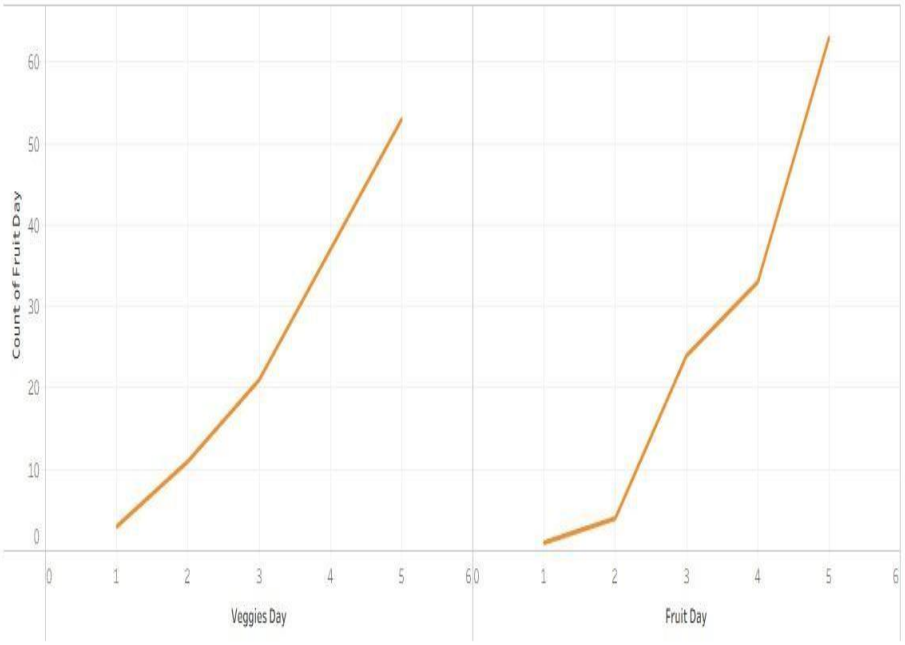
"It says about Meal payment Habits."

"This pie chart talks about physical fitness of people."

"It shows about how often students play sports."

"Daily consumption of fruits and vegetables is common. reflecting

"This b count male a



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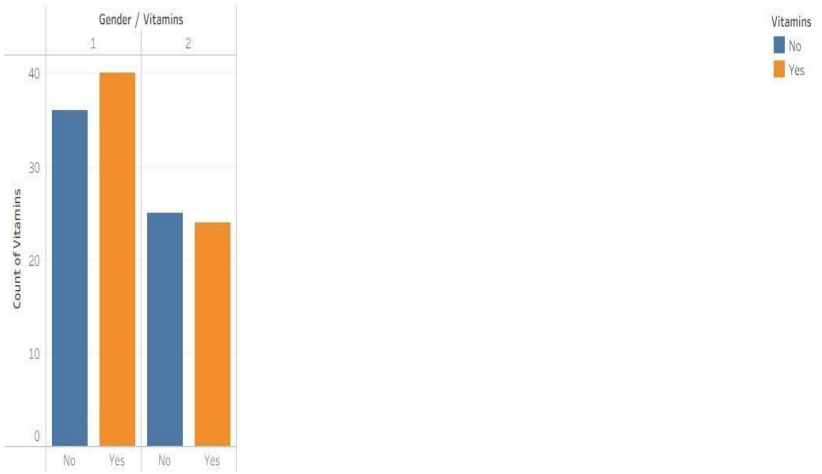
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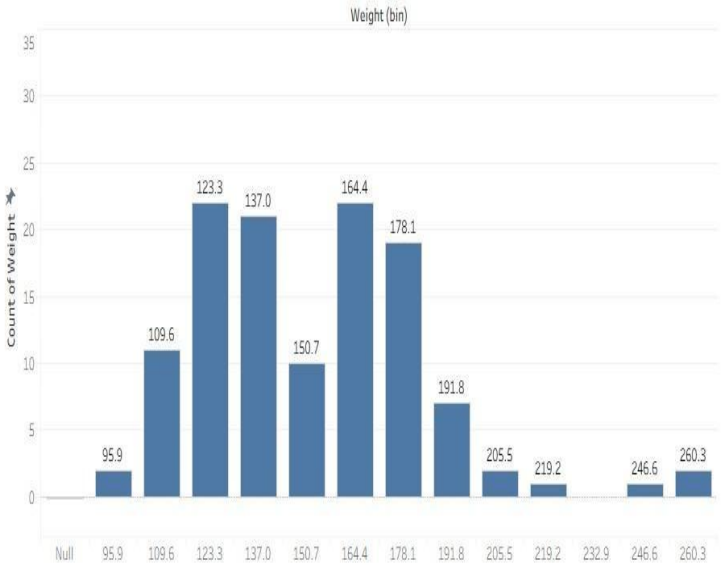
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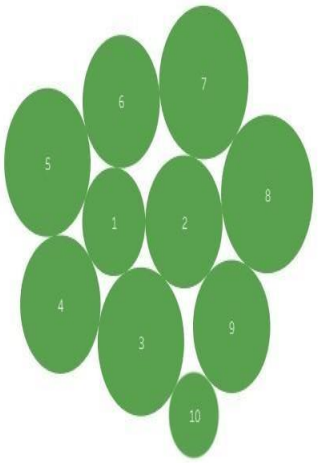
"This bar chart tells count of vitamins for male and female."

"This c weight



		majority of ents reported that parents cook 1-2 | "It says about Meal payment Habits." | "This pie chart talks about physical fitness of people." | "It shows about how often students play sports." | "Daily consumption of fruits and vegetables is common, reflecting | "This bar chart tells count of vitamins for male and female." | **"This chart is about weight distribtion."** | "Most healthy reward

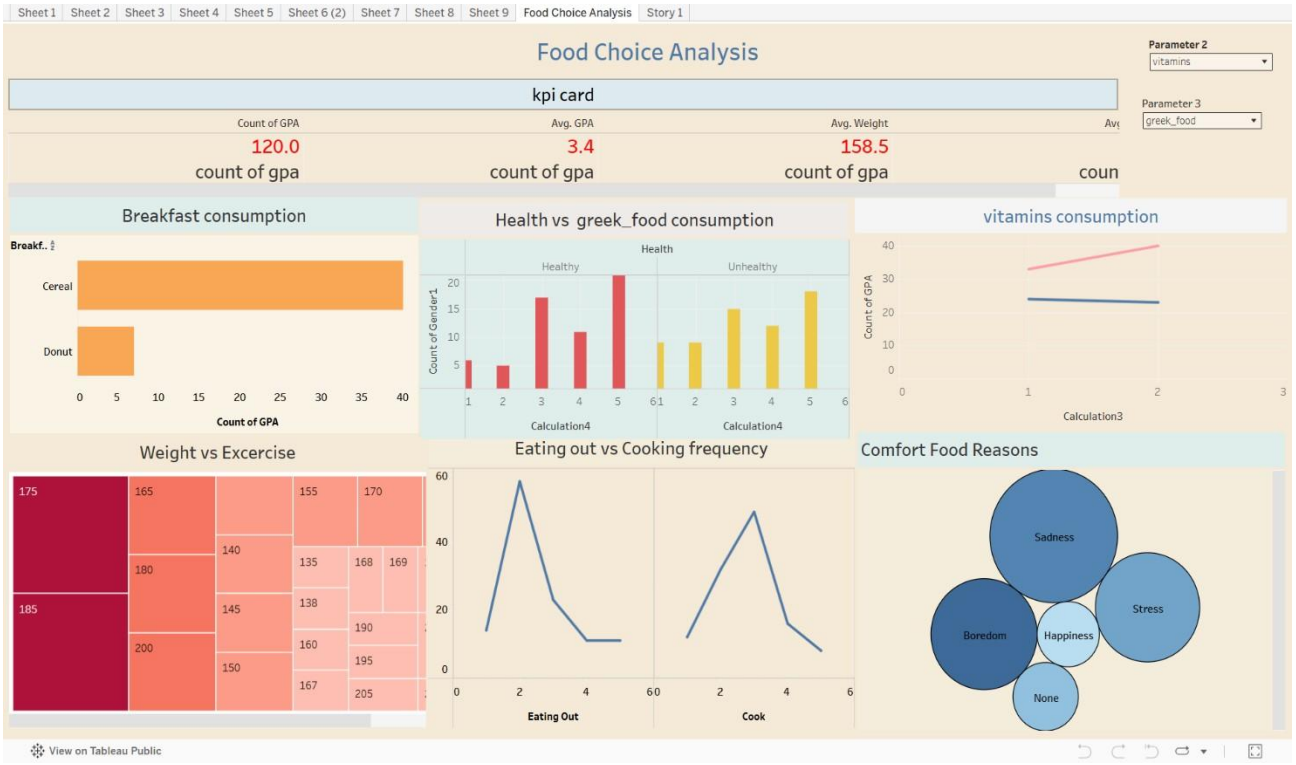


		t "It says about Meal payment Habits." "This pie chart talks about physical fitness of people." "It shows about how often students play sports." "Daily consumption of fruits and vegetables is common, reflecting "This bar chart tells count of vitamins for male and female." "This chart is about weight distribtion." "Most students find healthy eating to be a rewarding aspect of 

7. RESULTS

7.1 Output Screenshots

Dashboard:01



Dashboard link :

- https://public.tableau.com/shared/8M7NZRJ6C?:display_count=n&:origin=viz_share_link

Story links :

- https://public.tableau.com/shared/K24495CN3?:display_count=n&:origin=viz_share_link
- https://public.tableau.com/shared/75F3FXBXJ?:display_count=n&:origin=viz_share_link
- https://public.tableau.com/shared/5Q24J9KG2?:display_count=n&:origin=viz_share_link
- https://public.tableau.com/views/Food_Choice_Analysis1/Story1?:language=en-

US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8.ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Enables data-driven decision making for food choices	Data collection from cafeteria and students can be inconsistent
Promotes healthier eating habits among students	Potential privacy concerns with food consumption data
Uses Tableau for intuitive and interactive data visualization	Some users may resist using or learning the dashboard
Provides actionable insights to health coordinators and campus staff	Initial setup requires technical skills and effort
Supports personalized dietary suggestions	Requires ongoing maintenance and regular data updates
Educates students about nutrition and wellness	Limited impact if students do not engage with the tool
Scalable model for other educational institutions	Relies heavily on institutional support and collaboration

9.CONCLUSION

This project helps students choose healthier food using easy-to-understand Tableau dashboards.
It supports better eating habits and wellness on campus.
In the future, it can be improved and used in more colleges.

10.FUTURE SCOPE

- **Personalized Diet Plans** using AI/ML for student-specific recommendations
- **Mobile App** for real-time access and convenience
- **Gamification** to promote healthy eating habits
- **Behavior Tracking** through long-term analytics
- **Cafeteria Insights** for menu improvements
- **Fitness App Integration** for a holistic health profile
- **Admin Dashboards** for institutional health reporting
- **Multi-Campus Expansion** to replicate in other colleges
- **Live Feedback** from students on meals
- **Sustainability Tracking** via food waste and consumption data.

Source Code : NIL

Dataset link : https://www.kaggle.com/datasets/borapajo/food-choices?select=food_coded.csv

GitHub link:

Project Demo link: https://drive.google.com/file/d/1NggkIe-0RduZ4cPWDDS_AytT-Zdtad-S/view?usp=sharing